Systematic Literature Review on Technology-Based Fact Verification

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***Abstract*—Misinformation can lead to various adverse outcomes, including financial losses, misunderstandings, and even psychological issues, which can be mitigated through news fact verification. This research addresses a significant gap in the literature by conducting a systematic and comprehensive review of fact verification technologies in Indonesia. The literature review method in this study utilizes the Population, Intervention, Comparison, Outcomes, and Context (PICOC) framework to identify and evaluate research related to fact verification. We found that efforts to develop fact verification datasets in Indonesia remain scarce, with most existing datasets being in English. One study involved an Indonesian language dataset, namely XFEVER, which uses a translated Indonesian dataset for testing but was not included in the training process. The essential components of a fact verification dataset, such as claim, label, and evidence, must be accurately adapted to fit the context of Indonesian local news. Another finding is that while algorithms like RoBERTa and BERT are frequently used, they require further adaptation for the Indonesian context. This presents an opportunity for future research on training and testing fact verification datasets using multilingual languages, particularly Indonesian, to enhance digital literacy and information discernment skills among the Indonesian public.**

Keywords— fact verification, model, indonesian language

I. INTRODUCTION

Information obtained quickly can make people's lives easier [1], [2], but this also brings challenges due to ongoing technological advances. In the ever-growing digital era, the existence of accurate and trustworthy information is very crucial. Information plays an important role in decision-making [3], [4], both at the individual and professional levels. The truth of information becomes very important when people rely on news for important aspects such as health [5], education [6], and politics [7]. Incorrect or unreliable information can cause harm [8], misunderstanding [9], and psychological issues [10]. Therefore, it is important for information consumers to choose trustworthy sources, verify the truth before sharing it, and be active in disseminating accurate information. People who are skilled in managing information can more effectively combat the challenges of a digital world filled with fake news and misinformation.

In today's world, the reliance on knowledge and technology is unprecedented, necessitating a strong grasp of both by nearly everyone. As a result, enhancing information literacy is crucial to navigate the deluge of data effectively. The advent of social media and online news portals has transformed how information is shared, a development that carries both benefits and drawbacks. While these platforms facilitate the rapid dissemination of news and ideas, they also contribute to the proliferation of fake news, including clickbait, hoaxes, propaganda, and rumors [11], [12]. Unfortunately, the speed of spreading information often overshadows the accuracy, posing significant challenges in discerning truth from falsehood [13], [14]. It is vital for users of social media and online news portals to critically evaluate the credibility of sources, engage in fact-checking, and support initiatives aimed at curtailing the spread of misinformation. By adopting these practices, information consumers can mitigate the adverse impacts of this issue and contribute to the promotion of higher-quality information.

Currently, there are still limitations in the datasets used for fact verification on news portal sources. Most research in this area relies on encyclopedia sites like Wikipedia [15], [16], [17]. However, the use of Wikipedia is often critiqued due to its public editing feature, which may lead to inaccuracies or bias. To address these issues, a methodology is needed to verify the accuracy and objectivity of information from various sources, including news portals. The difference is that fact verification for news portals is conducted based on news sources that are currently developing in a specific region, area, or country, whereas encyclopedia verification tends to feature news that occurred in the past over a longer period compared to news from portals. In fact, news portals are often used as a reference for news in encyclopedias. This requirement aligns with advancements in machine learning and natural language processing techniques, which can automatically sort and validate information. Consequently, the developed technology can facilitate stakeholder engagement in the fact verification process, thereby enhancing awareness and digital literacy skills in information evaluation. Additionally, in the context of fact verification in the Indonesian language, there are still technological limitations. The Indonesian fact verification dataset from the XFEVER translation was not fully utilized during the model implementation process. XFEVER in Indonesian was only involved in the testing process, which means the model does not understand words or sentences from news or claims that use idioms, cultural expressions, or informal language different from the language used during training. Consequently, models that are not trained in the target language increase the risk of misinterpreting the diverse vocabulary of the Indonesian language. However, the challenges of Indonesian fact verification datasets are still limited compared to English fact verification datasets, which

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have undergone extensive training and testing in existing research. [15], [18].

This research aims to enhance the literature by providing systematic and in-depth observations on fact verification technology in Indonesia. It focuses on analyzing technologies previously utilized in various countries and their adaptation for Indonesia, offering valuable guidance for policymakers, media practitioners, and technology developers. The contributions of this research include:

- A comprehensive analysis of fact verification technology, with a particular emphasis on datasets,
- How does the model interpret fact verification dataset annotations?
- What are the popular algorithms used for optimizing the classification process?
- How is the performance of the algorithms in addressing the challenges of fact verification?

This paper is organized into several sections: Section 2 outlines the research methodology; Section 3 presents a discussion and the results from the literature review; Section 5 concludes with insights derived from the literature review regarding the implementation of fact verification.

II. Research Method

The methodology used in this research is the Semantic Literature Review (SLR) to assess literature related to fact verification. This research process consists of several stages starting with research design, namely compiling a PICOC, which is then used to determine research questions. Next, the search strategy and selection criteria were determined, followed by data extraction. The final stage was data synthesis, which resulted in a summary visualization of the literature study. The method used in this research can be seen in Figure 1.

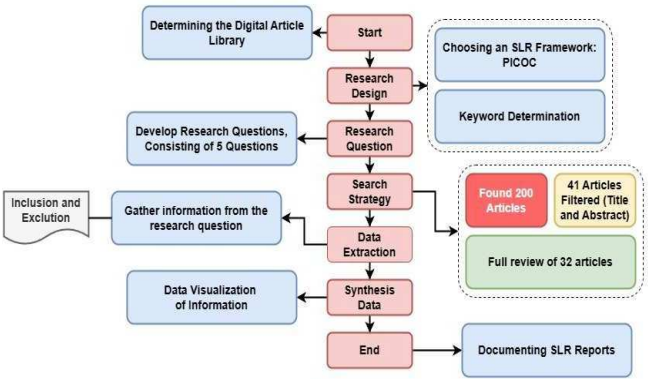


Fig. 1. Overview of fact verification SLR research

A. Research Design

The research design in this study utilizes the Population, Intervention, Comparison, Outcomes, and Context (PICOC) framework as a systematic approach to identify, analyze, and evaluate studies related to fact verification. This framework ensures a structured and comprehensive review of the literature, allowing the key elements of each study to be examined in detail. Details of the compiled PICOC can be seen in Table 1.

TABLE I. PICOC Structure

No	PICOC	Definition	Keyword
1	Population	Dataset,Fact Verification	Fact Verification
2	Intervention	Algorithms used for fact verification	Algorithm, Method, Model
3	Comparison	Performance of machine learning algorithms, comparing datasets before and after the intervention	Algorithm Performance
4	Outcomes	Accuracy of the algorithm in classifying facts or effectiveness of the developed dataset	-
5	Context	Journals published at the latest in 2020, News Portals	-

B. Research Question

Determining research questions is a crucial stage in planning a systematic literature review because it guides the entire research process. These questions are prepared based on the PICOC criteria, which include Population, Intervention, Comparison, Outcome, and Context. The formulation of research questions that have been developed using the PICOC strategy is explained in Table II. The research questions used in this review are listed in Table II, according to the formulation in Table I.

TABLE II. Research Question

Code	Question
RQ1	What dataset is used for fact verification?
RQ2	Does it use manual or automatic annotators in the dataset creation?
RQ3	What algorithm method is most popular in comparing and optimizing the text classification process?
RQ4	How does the performance comparison between algorithms in fact verification task?

C. Search Strategy

In the research strategy phase, articles are identified using specific, predetermined keywords. For this study, the selected digital libraries include IEEE Xplore, ACM Digital Library, ScienceDirect, and Web of Science. Initially, 200 articles were gathered. These were then narrowed down to 70 articles after reviewing titles and abstracts for relevance to the research topic.

Further scrutiny of the full content of these journals resulted in a refined selection of 57 articles. Table III presents a comprehensive list of the digital libraries utilized for this systematic literature review.

TABLE III. Digital Library Tables

Library	Definition
IEEE Xplore	https://ieeexplore.ieee.org/
ACM Digital Library	https://dl.acm.org/
ScienceDirect	https://www.sciencedirect.com/
Web of Science	https://access.clarivate.com/

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D. Selection Criteria

The selection criteria applied in preparing the Systematic Literature Review (SLR) are designed to guarantee the relevance and quality of the articles analyzed. Additionally, these criteria help identify the novelty of the dataset and the fact verification methods that are suitable for adaptation into an Indonesian language dataset. The criteria for article selection were based on the relevance of the topic to fact verification, the use of scientifically valid methods, the use of fact verification datasets, as well as articles published in reputable peer-reviewed journals. Articles that did not meet quality standards or were not directly related to the topic of fact verification were excluded from the review. The details of our inclusions and exclusions are outlined in Table IV.

TABLE IV. TABLE OF INCLUSION AND EXCLUSION CRITERIA

Inclusion	Exclusion
Literature using Deep Learning algorithm	Not using deep learning algorithm
Using a fact verification dataset	Not using a fact verification dataset
Model accuracy against Dataset	No Model accuracy against Dataset

E. Data Extraction

After completing one stage of the selection criteria, the next step involves extracting data from articles that meet these criteria. At this stage, we categorize the identified articles based on the research questions. These categories have been devised to meet the needs of our future research and to help us create a new dataset in Indonesian. The categories, derived from the research questions, are outlined in Table V.

TABLE V. DATA EXTRACTION CATEGORIES TABLE

Research Question	Category
RQ1	Dataset
RQ2	Annotation Method
RQ3	Algorithm
RQ4	Algorithm Performance

F. Sintesis Data

Data synthesis involves analyzing and integrating information from extracted data to identify trends. This process informs future research decisions related to developing and testing datasets for fact verification in Indonesian. At this stage, we present the gathered information through narratives and visualizations to enhance understanding and accessibility.

III. RESULT AND DISCUSSION

Our literature review revealed that most studies broadly address fact verification without focusing on specific subjects, except for the works cited in [19], [20], [21] , and [22], which concentrate on health-related topics. Furthermore, we observed that the development of datasets is relatively uncommon, and those that are developed predominantly use English. This observation highlights a unique challenge for Indonesian contexts, particularly in dataset development. Therefore, refining the sentence structure in Indonesian for Natural Language Processing (NLP) prior to model implementation presents a promising avenue for future research. Compared to other studies focusing on fact verification literature reviews, such as the one conducted by Yang et al. [23], this research focuses on filtering information related to multilingual fact verification technologies, specifically Indonesian.

A. What Dataset is Used for Fact Verification?

In this literature review collection activity, we obtained various types of datasets. Of the 42 works of literature that we collected, the most frequently used dataset is FEVER, as shown in Figure 2, with 10 articles [24], [25], [26], [27], [28], [29], [30], [31], [32], [32] using this dataset. The FEVER dataset architecture has the potential to be adapted into Indonesian, with several adjustments that utilize Natural Language Processing (NLP). We present a summary of the results of the grouping of the data we collected in Figure 2.

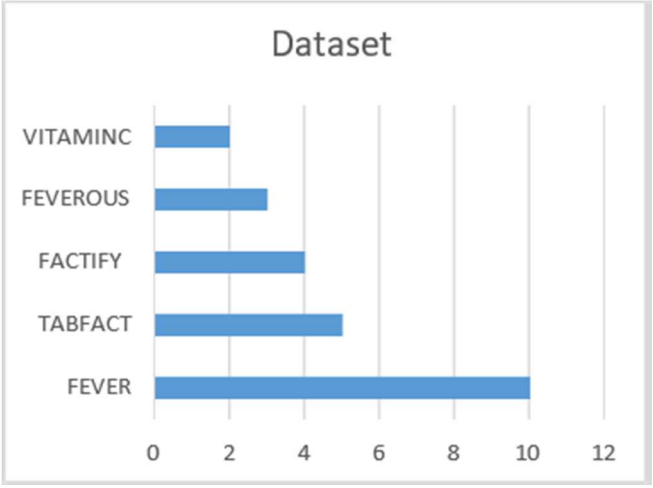


Fig. 2. Most used dataset in fact verification research

On the other hand, we found that most datasets in fact verification research use sources from Wikipedia. Only a few studies involve news portals in creating datasets. Based on our observations, news portals are faster at creating news compared to Wikipedia, which tends to update information over a longer timescale. Therefore, fact verification datasets that include multiple sources from news portals can provide an edge in terms of the timeliness and relevance of information. Data integration from news portals will increase the coverage and diversification of content and strengthen the accuracy and responsiveness of the verification system to the latest news. These findings recommend using the FEVER dataset architecture to develop an Indonesian language dataset consisting of labels, claims, and evidence sourced from news portals.

B. Does it Use Manual or Automatic Annotators in the Dataset Creation?

Based on the results of the literature study, we found the FEVER dataset at [24], [25], [33], [26], [27], [28], [29], [30], [31], [32], TABFACT at [34], [35], [36], [37], [38]. FACTIFY at [39], [40], [41], [42], FEVEROUS at [39] [43], [38], VITAMINC at [44], [32], which uses a manual annotator. Dataset [45],[46] uses a semi-automatic annotator. Dataset [47], [48] uses an Automatic annotator. For a tabular representation of the distribution of annotation methods, refer to Table VI.

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TABLE VI. ANNOTATOR IN THE DATABASE

Reference	Annotation in the Dataset
[24], [25], [33], [26], [27], [28], [29], [30], [31], [32], [34], [35], [36], [37], [38], [39], [40], [41], [42], [39] [43], [38]	Manual Annotator
[45], [46]	Semi-Automatic
[47], [48]	Automatic

Manual annotation, performed by humans, may introduce subjectivity into the annotations. To address this issue, we propose conducting comparative tests between manual and automatic annotation methods to assess their objectivity in sentence annotation. One effective technique for automatic annotation is Named Entity Recognition (NER) [49]. Our recommendation is to test manual, semi-automatic, and automatic annotators on the Indonesian language dataset when implemented in deep learning models to identify the best annotation method.

C. *What Algorithm Method is Most Popular in Comparing and Optimizing the Text Classification Process?*

In our literature study, we found that the algorithm implemented in this article is popular, as evidenced by its success in research. The most common algorithms in the literature we reviewed [50], [41], [32], [38] are RoBERTa and BERT, which are part of the transformer algorithm family. Neural Network [25], [29], Ensemble [39], [41], Multimodal [51], [42], Logistic Regression [52], [53], Support Vector Machine [52], [53] is also an algorithm used for classification trained using vectorized data and output from the language model, with the aim of determining the validity of claims by analyzing the evidence provided and assigning appropriate labels.

One of the main challenges in verifying facts in Indonesian is adapting these algorithms to carry out the fact verification process in Indonesian. Apart from that, the Ensemble method is also of concern to us as a new challenge in developing Indonesian language fact verification models. In future research, it is hoped that RoBERTa and BERT will enable the use of the Ensemble method for verifying Indonesian language facts. In this study, we categorized the results of the review of popular models used, which are visualized in Figure 3.

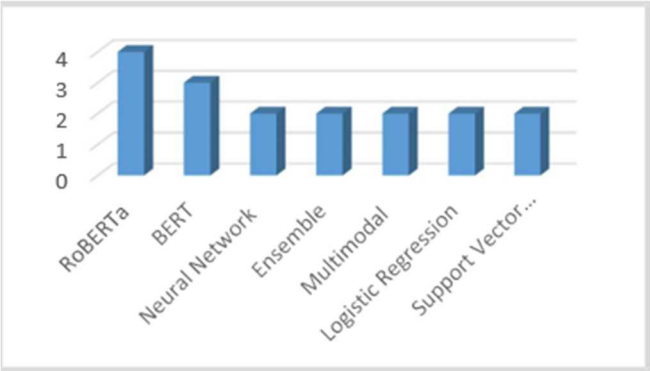


Fig. 3. Algorithm method is most popular

D. *What is the Evaluation Metrics of Fact Verification Task?*

From the results of our literature study, we identified that accuracy, precision, recall, F1 Score, and Dataset Score [51], [54] are essential indicators for evaluating and comparing the performance of fact verification models. These metrics provide a comprehensive framework to assess how well different models perform in handling fact verification tasks. Furthermore, these findings serve as a valuable reference for testing Indonesian language verification datasets across various algorithms. By doing so, researchers can identify the most effective algorithms tailored to the specific challenges of verifying facts in the Indonesian context, ensuring higher accuracy and reliability in real-world applications.

IV. CONCLUSION

This study conducts an in-depth analysis of fact verification technology, emphasizing dataset usage and the development of machine learning. The literature reveals that most fact verification research does not concentrate on specific topics, although there are notable exceptions in health-related studies. Notably, the development of datasets tailored for Indonesia remains scarce, with a predominance of English-based datasets, posing challenges in adapting sentence structures and NLP for Indonesian fact verification models. The primary dataset used in existing research is FEVER, primarily sourced from Wikipedia. However, the utilization of news portal sources is limited, despite their potential to enhance the timeliness and relevance of information. Typically, datasets comprise three main labels: claim, label, and evidence, which must be accurately adapted for the local Indonesian news context. Dataset creation often relies on manual annotators rather than automatic methods, such as Named Entity Recognition (NER), which could enhance objectivity. Transformer algorithms like RoBERTa and BERT are frequently used, yet there remains a need for their adaptation to Indonesian, alongside opportunities to explore Ensemble methods. The effectiveness of these algorithms in fact verification tasks is evaluated using metrics like accuracy, precision, recall, and F1 Score, providing crucial benchmarks for developing a more accurate and relevant fact verification system in Indonesia. These insights lay a solid groundwork for future research aimed at advancing fact verification technology in Indonesia.

Suggestions for future work include the development of an automatic fact verification system utilizing pre-trained models that have already been tested and proven to deliver strong performance. By leveraging pre-trained models, researchers can benefit from existing knowledge and efficiencies, significantly reducing the need for extensive training from scratch. Additionally, these models can be fine- tuned and adapted to handle specific challenges related to fact verification, such as the complexities of language and context. Future systems should also focus on improving accuracy and scalability, ensuring they can process large volumes of data across multiple domains. This approach has the potential to enhance the reliability and speed of fact verification in real-world applications.

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